Quang-Vinh Dang
British University Vietnam
Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

To the Editors,
Computer Vision and Image Understanding

Dear Editors,

I am submitting the manuscript "WAGER: Within-cell Antisymmetric Gain Evaluation of Reasoning" for consideration as a regular paper in *Computer Vision and Image Understanding*.

The paper concerns how progress is measured on vision benchmarks whose annotations carry a strong prior. In Visual Genome scene graph generation, the ordered object-class pair predicts the relation label well enough that a model can raise its headline score by fitting that annotation regularity more precisely, without becoming any better at reading the image. The standard audit compares a model against a frequency baseline, but that cannot answer the question a benchmark result actually poses: when system B outscores system A, how much of the gain came from the image rather than from the prior?

WAGER answers that question directly. Given two frozen probabilistic models, it forms their proper-score contrast, matches test examples that share a declared prior feature, and crosses their labels. This yields an exact finite-sample identity — the observed gain equals a prior-transported gain plus an antisymmetric instance-alignment gain — in which the residual is an order-two U-statistic that, under the quadratic score, equals the within-cell prediction–label covariance the new model has gained. The estimator runs in $O(NK)$ time on cached predictions, fits no baseline model, and comes with image-cluster-robust intervals and a cell-stratified randomization test. Two further exact results are proved: leave-one-out transport removes an attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in incurs, and coarsening the audit cell shifts the estimate by an explicit between-cell covariance term.

The empirical findings are, I believe, of direct interest to this readership. On 229,605 VG150 PredCls relations, a class-only network's entire improvement over the frequency baseline is provably prior-transported, with exactly zero instance alignment. More strikingly, when a frozen CLIP encoder replaces annotation-derived box geometry, the resulting model scores 0.04655 *lower* and is three accuracy points worse — an evaluator reading either number would discard it — yet it is shown to align with individual instances significantly *better* than the model that beats it (+0.00641, $p = .002$), its entire deficit lying in the annotation prior. Applying the same estimator unchanged to long-tailed classification reproduces the pathology from the opposite side: a re-weighting schedule whose near-zero aggregate gain conceals a +0.197 prior improvement cancelling a −0.196 alignment loss. In both cases the aggregate score is not merely imprecise but actively misleading about what the benchmark is meant to measure.

I believe the work fits the journal's scope under Theory, and speaks to matching and recognition through what it establishes about a widely used vision benchmark. It is a measurement contribution rather than a new recognition architecture: it makes no state-of-the-art claim, and its value lies in an exact identity, the inference built on it, and the concrete findings about VG150 and CLIP features reported above. All code, cached model outputs and scripts needed to reproduce every number and figure are publicly available, and the repository URL is given on the title page.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with the journal's policy, the manuscript includes a declaration of generative AI use in its preparation. Author details are supplied on the separate title page, with the manuscript itself anonymized for the double-anonymized review process.

Thank you for your consideration.

Yours sincerely,

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam